Introduction to Data Science

Visualization Design Principles

Cognitive Load

- Cognitive load is the amount of mental effort required to interpret information.
- Every single element you add takes up cognitive load.
- The goal in data visualization is to minimize cognitive load yet accurately communicate your message.

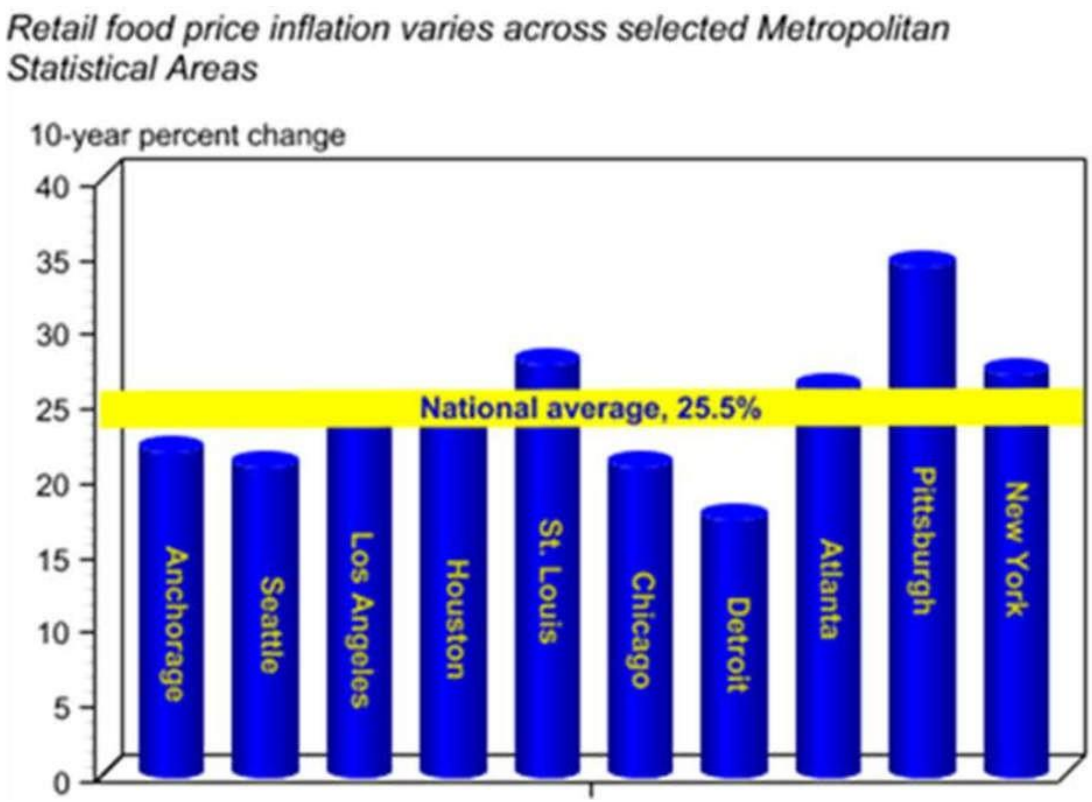
Clutter

- Clutter is all the things you can remove while still preserving key ideas.

- Reduce clutter to minimize user's cognitive load.

- Less clutter = more effective visualizations

Example



Retail food price inflation varies across selected Metropolitan Statistical Areas
10-year percent change
40
35
30
25
20
15
10
5
0
National average, 25.5%
Anchorage
Seattle
Los Angeles
Houston
St. Louis
Chicago
Detroit
Atlanta
Pittsburgh
New York

Gestalt Principles

Gestalt Principles

- Gestalt Principles of Visual Perception identify which elements in our visuals are signal (the information we want to communicate) and which might be noise (clutter).

1) Proximity
2) Similarity
3) Enclosure
4) Closure
5) Continuity
6) Connection

Proximity

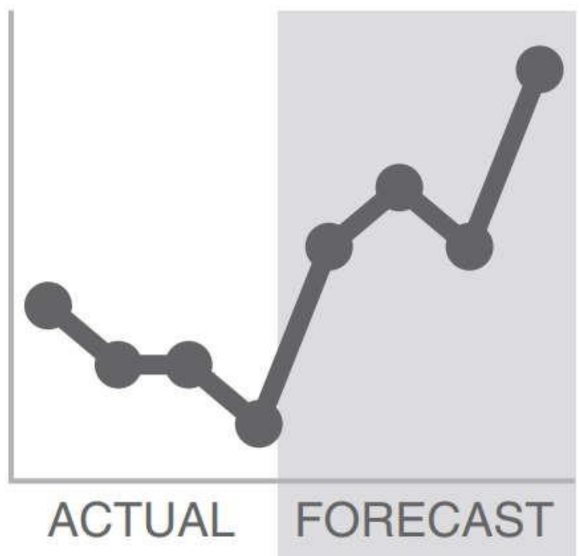
Application of proximity

Similarity

Application of similarity

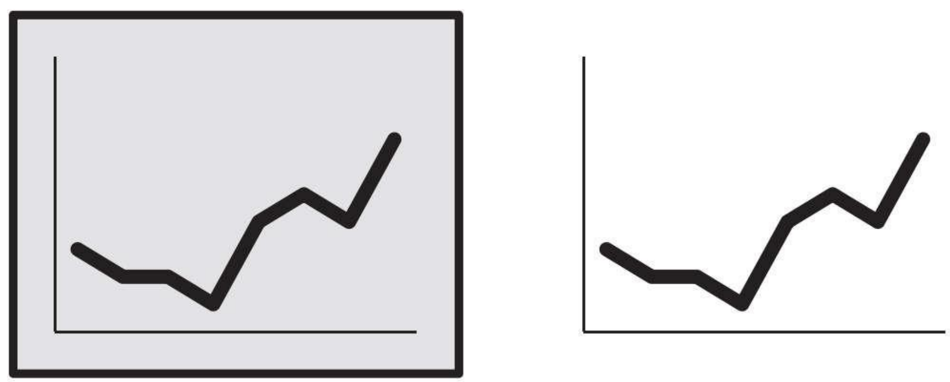
Enclosure

Application of enclosure

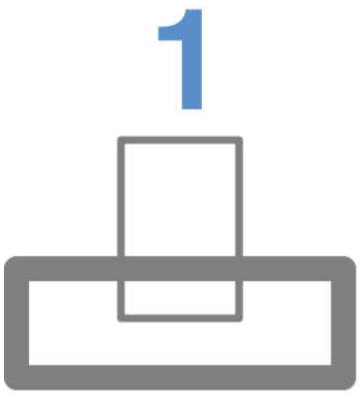


Closure

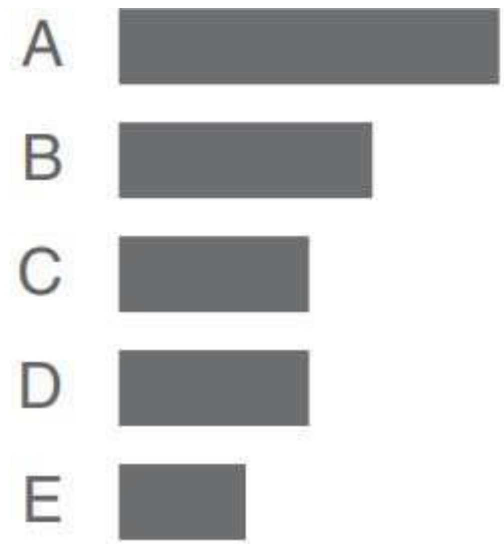
Application of closure



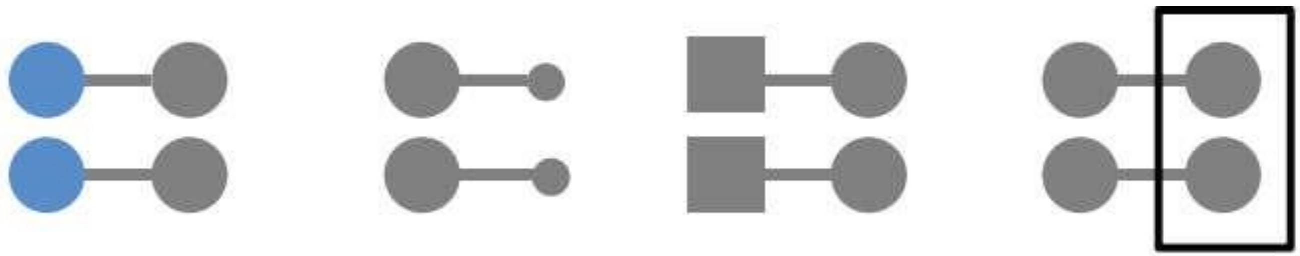
Continuity



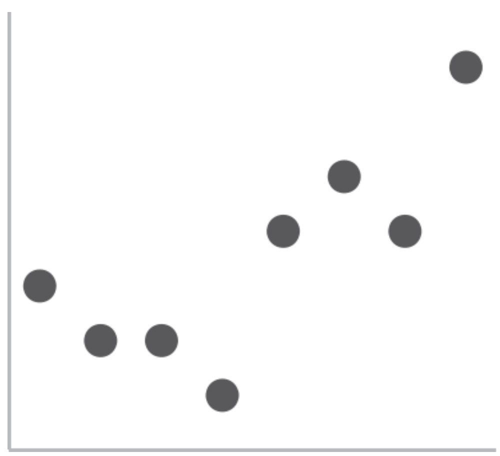
Application of continuity



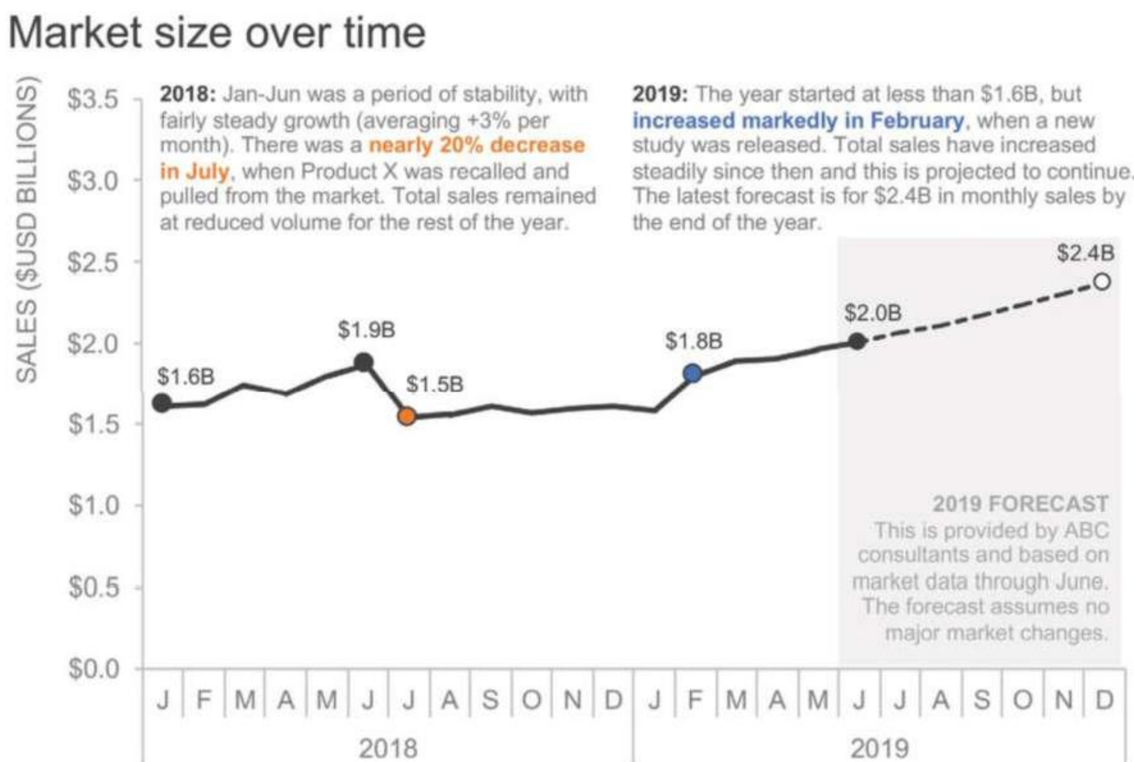
Connection



Application of connection



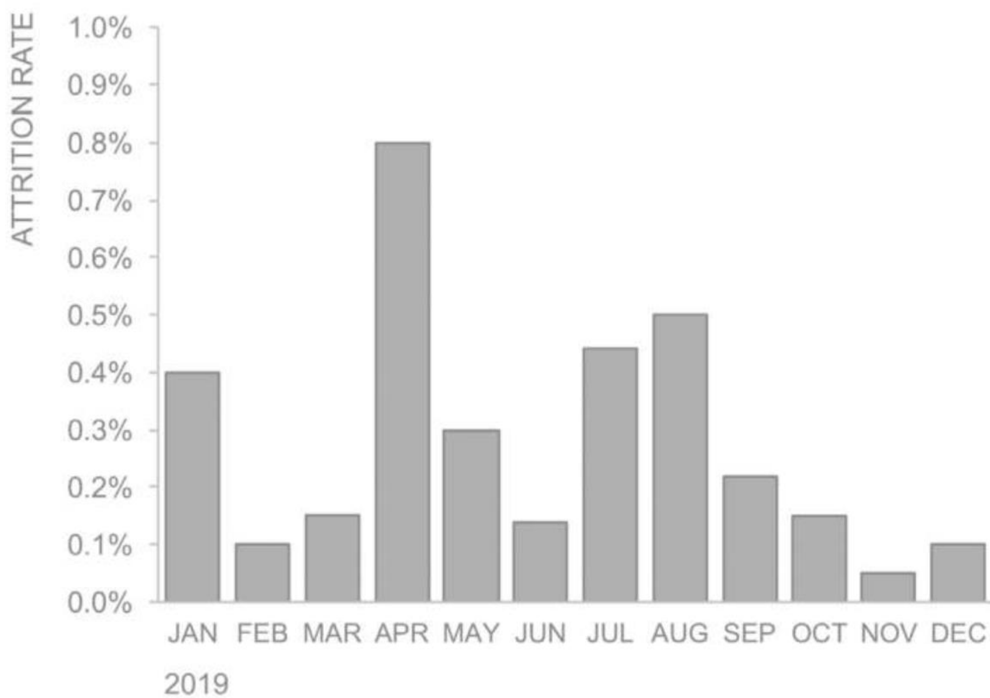
Which gestalt principles are in play?



Tying Words to Graph

Visually tie the words to the graph

2019 monthly voluntary attrition rate



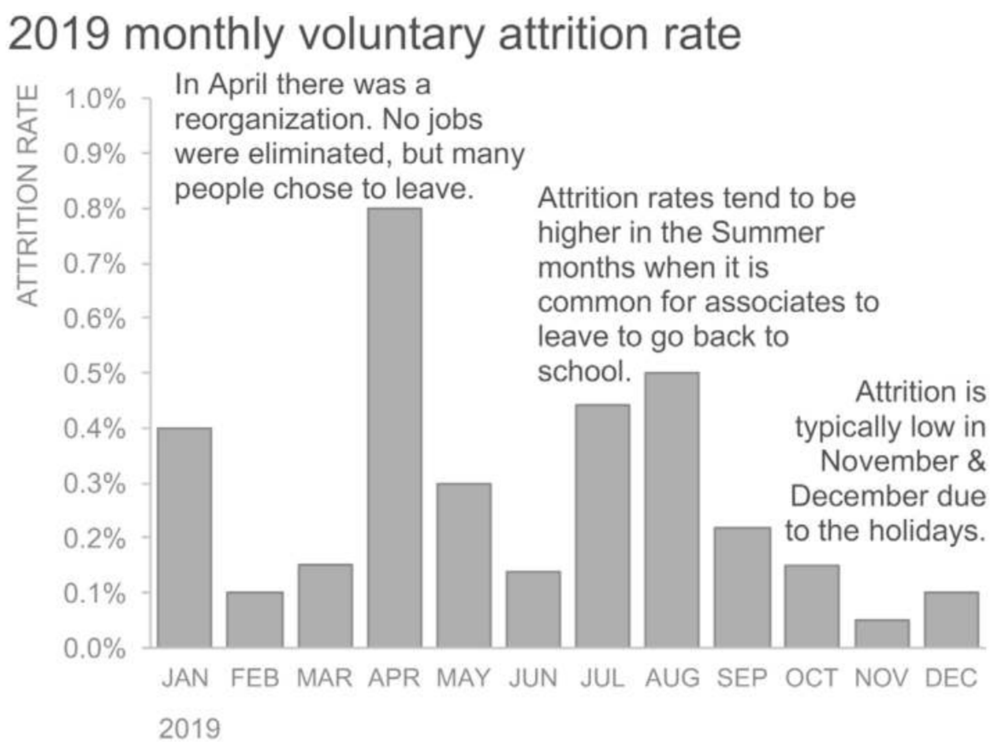
Highlights:

In April there was a reorganization. No jobs were eliminated, but many people chose to leave.

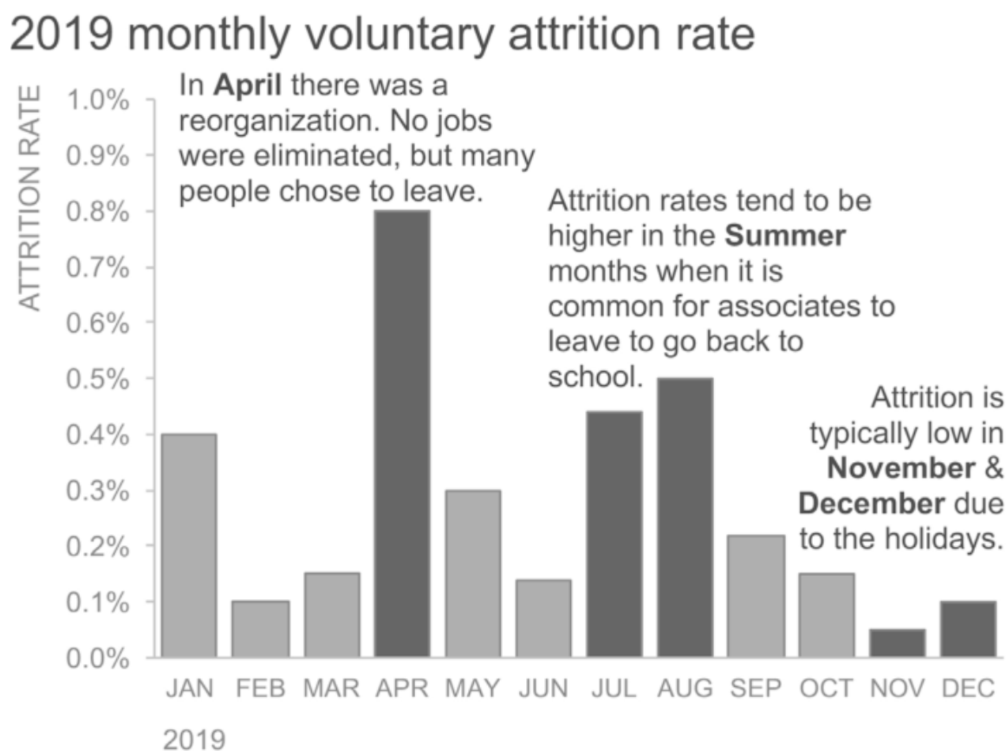
Attrition rates tend to be higher in the Summer months when it is common for associates to leave to go back to school.

Attrition is typically low in November and December due to the holidays.

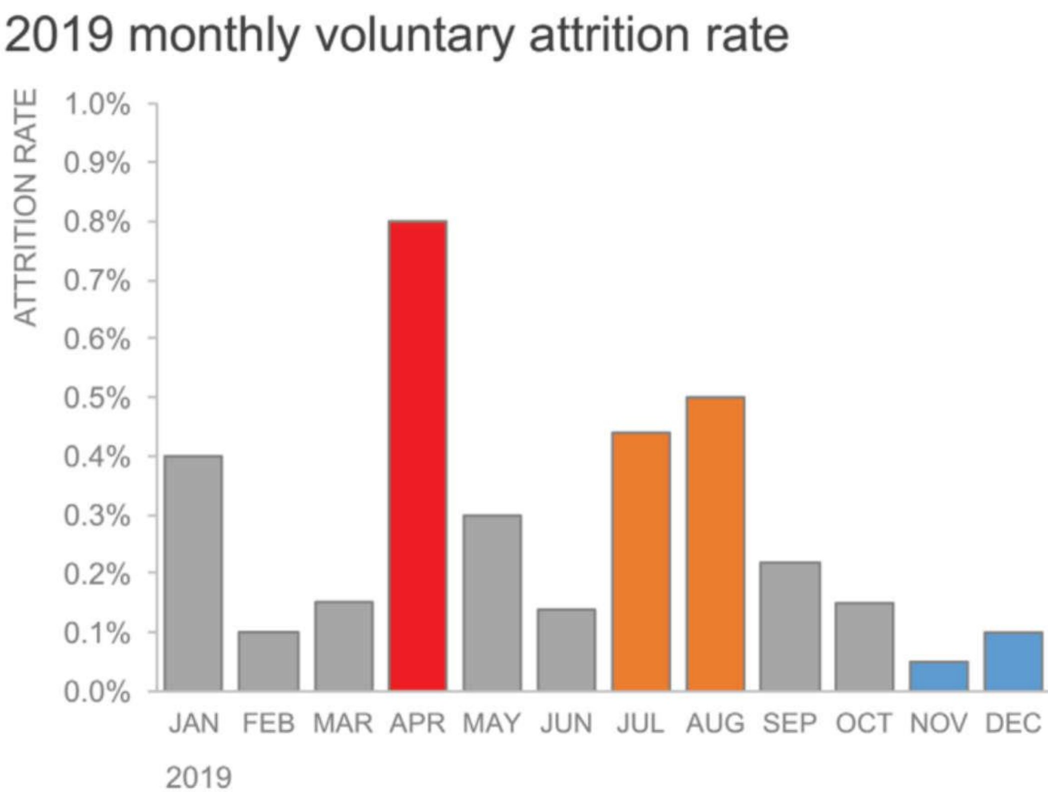
Proximity


2019 monthly voluntary attrition rate
ATTRITION RATE
1.0%
0.9%
0.8%
0.7%
0.6%
0.5%
0.4%
0.3%
0.2%
0.1%
0.0%
In April there was a reorganization. No jobs were eliminated, but many people chose to leave.
Attrition rates tend to be higher in the Summer months when it is common for associates to leave to go back to school.
Attrition is typically low in November & December due to the holidays.
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
2019

Proximity with emphasis



Similarity



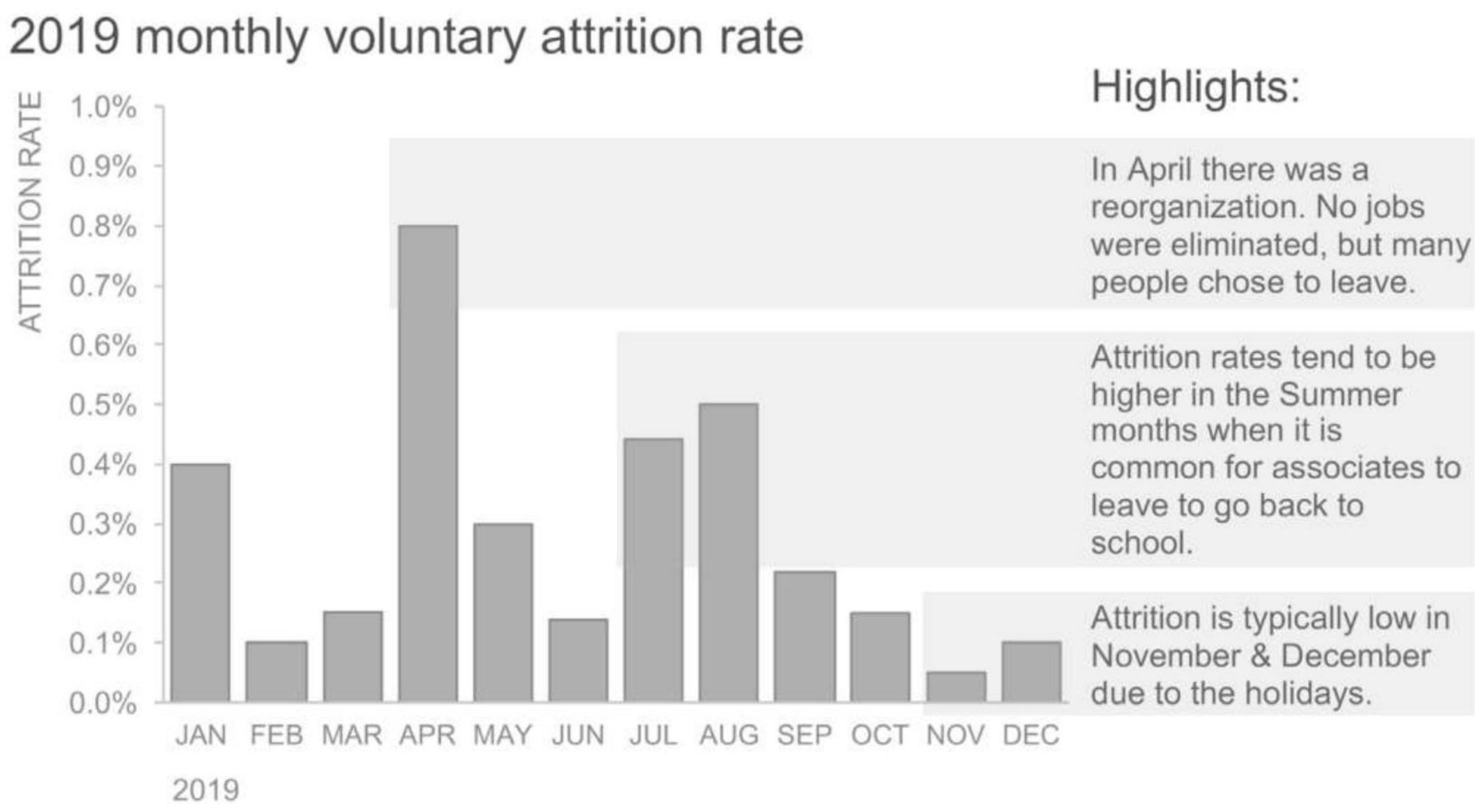
Highlights:

In **April** there was a reorganization. No jobs were eliminated, but many people chose to leave.

Attrition rates tend to be higher in the **Summer** months when it is common for associates to leave to go back to school.

Attrition is typically low in **November** & **December** due to the holidays.

Enclosure


2019 monthly voluntary attrition rate
ATTRITION RATE
1.0%
0.9%
0.8%
0.7%
0.6%
0.5%
0.4%
0.3%
0.2%
0.1%
0.0%
JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
2019
Highlights:
In April there was a reorganization. No jobs were eliminated, but many people chose to leave.
Attrition rates tend to be higher in the Summer months when it is common for associates to leave to go back to school.
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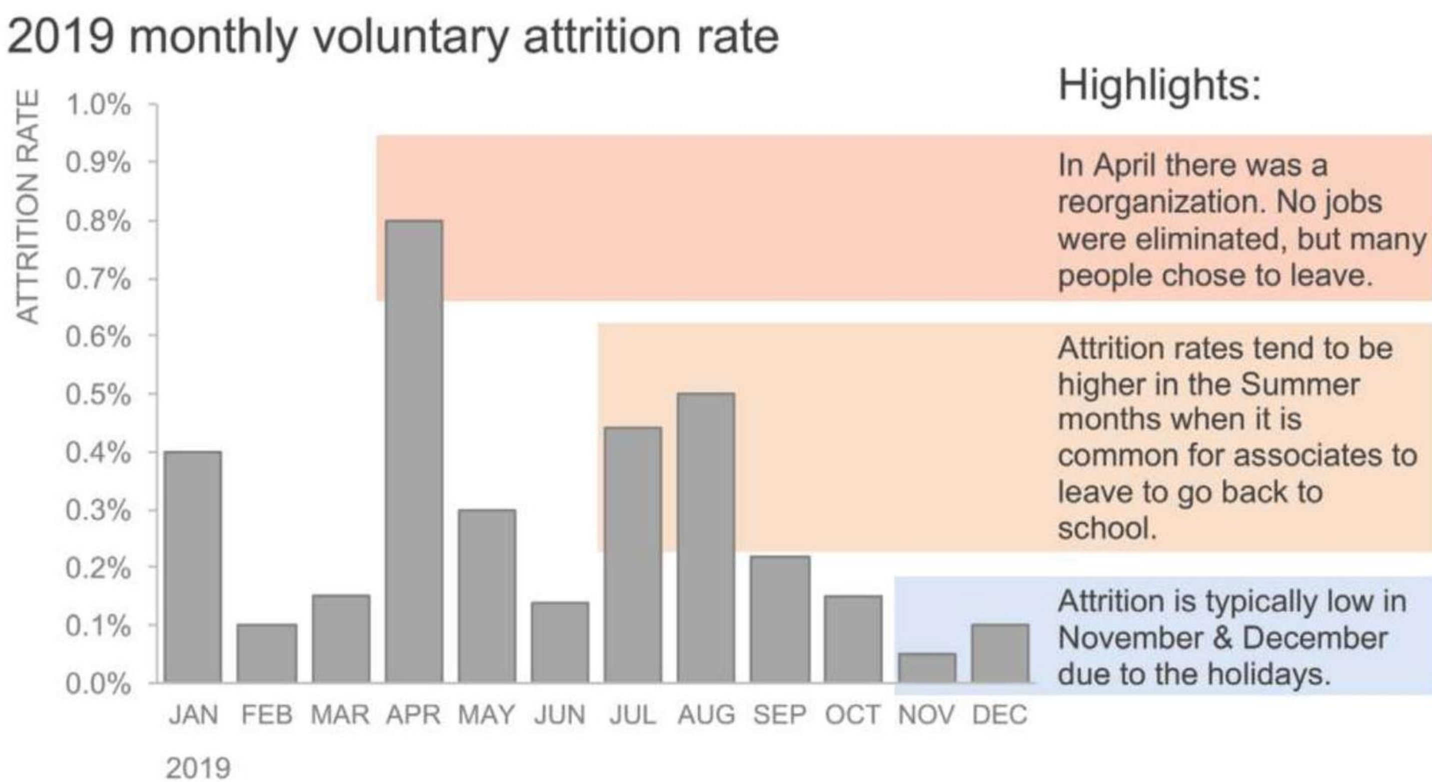
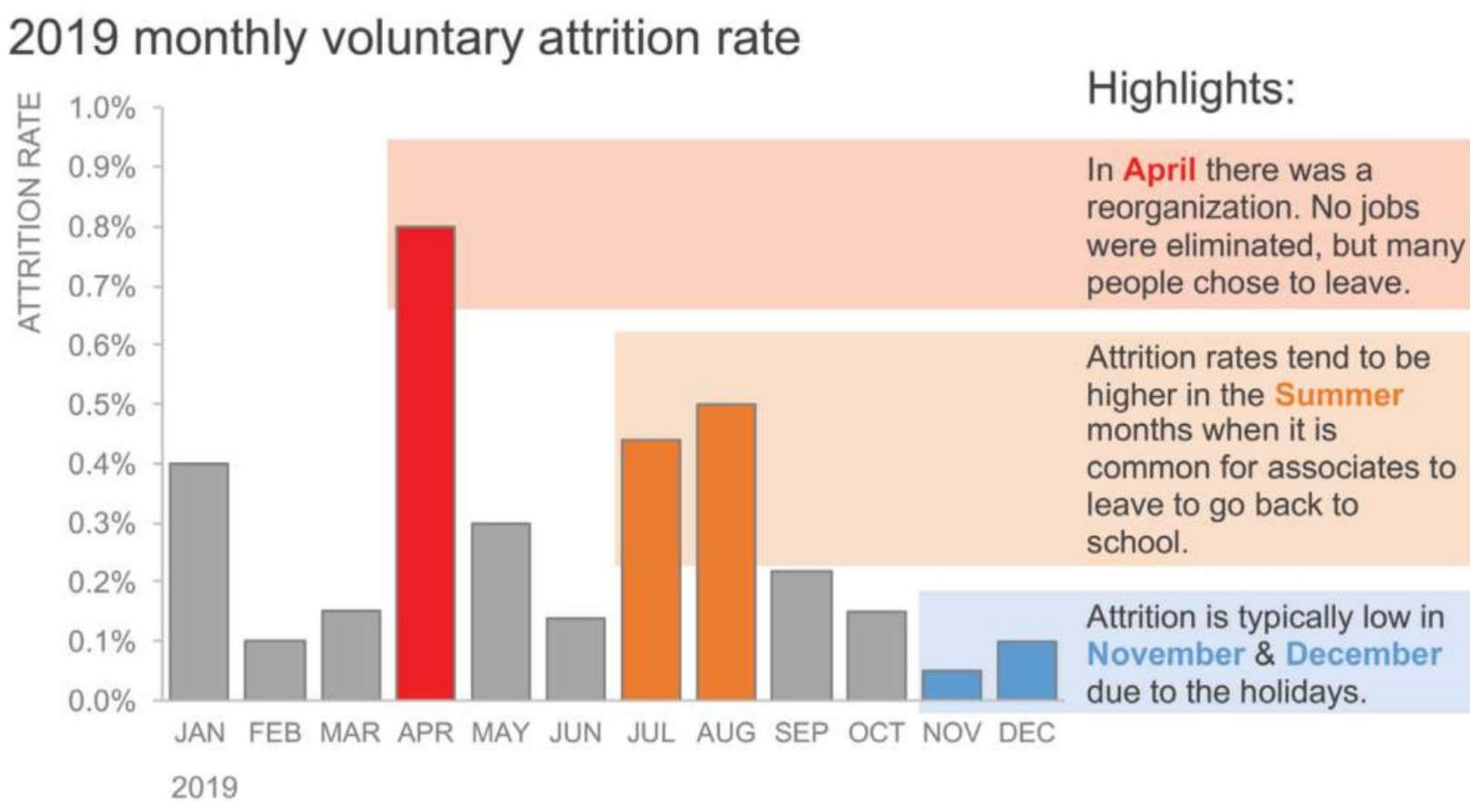
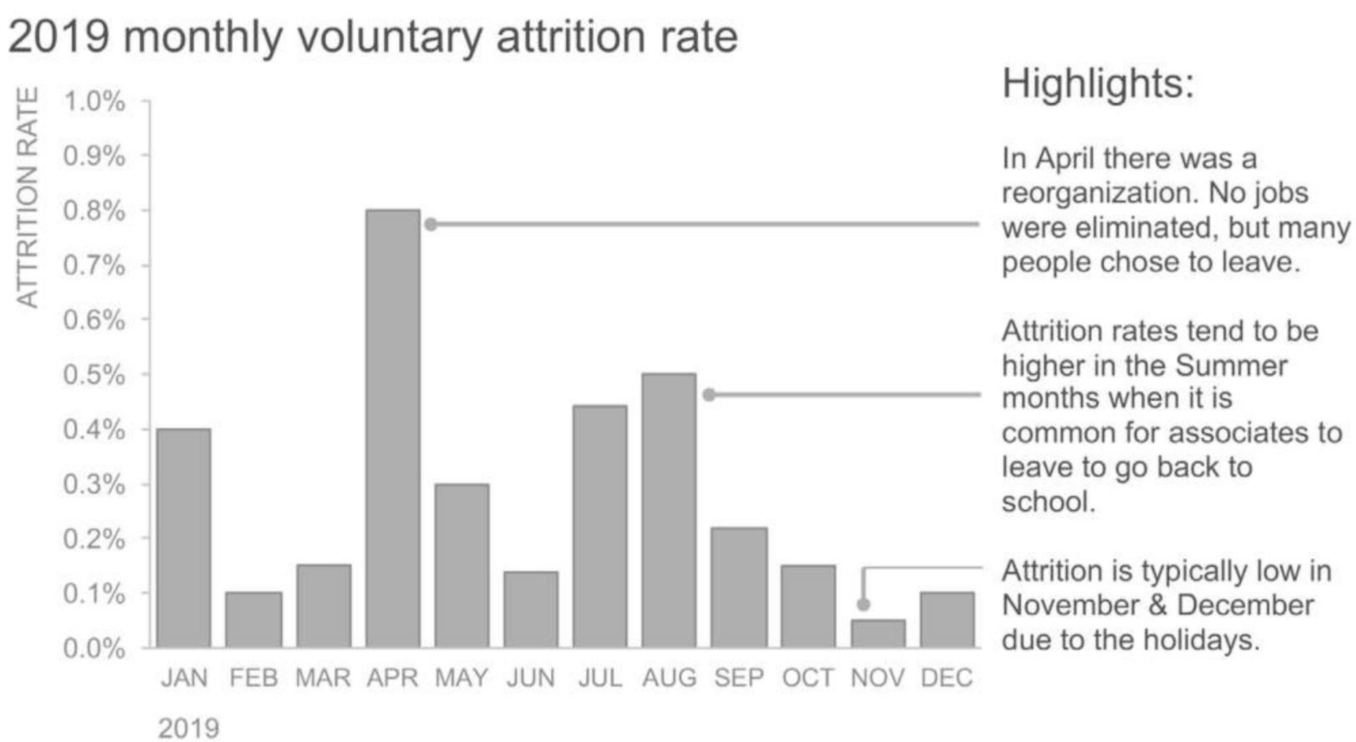
Enclosure with color differentiation



Enclosure + similarity



Connection


2019 monthly voluntary attrition rate
ATTRITION RATE
1.0%
0.9%
0.8%
0.7%
0.6%
0.5%
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0.1%
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JAN FEB MAR APR MAY JUN JUL AUG SEP OCT NOV DEC
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In April there was a reorganization. No jobs were eliminated, but many people chose to leave.
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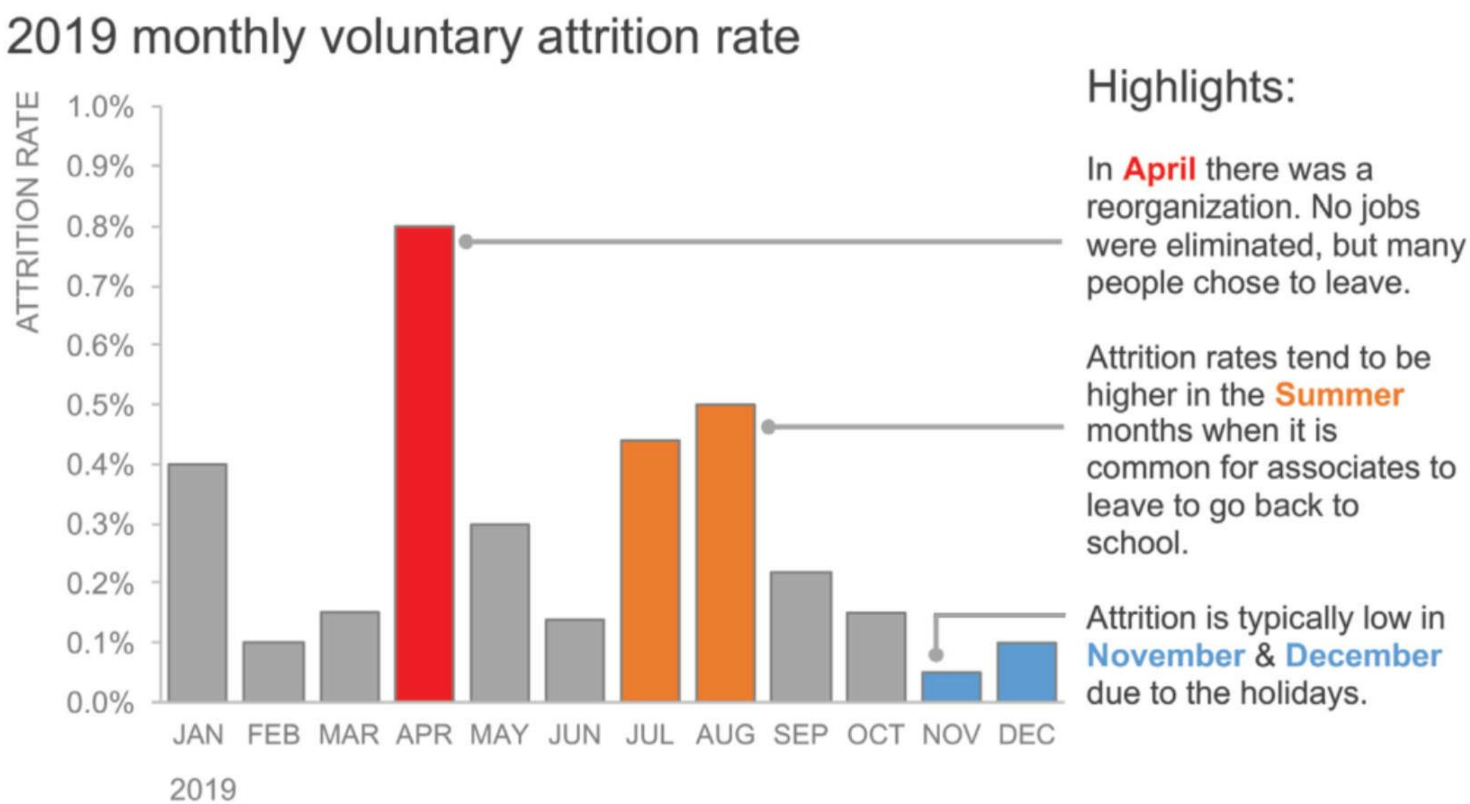
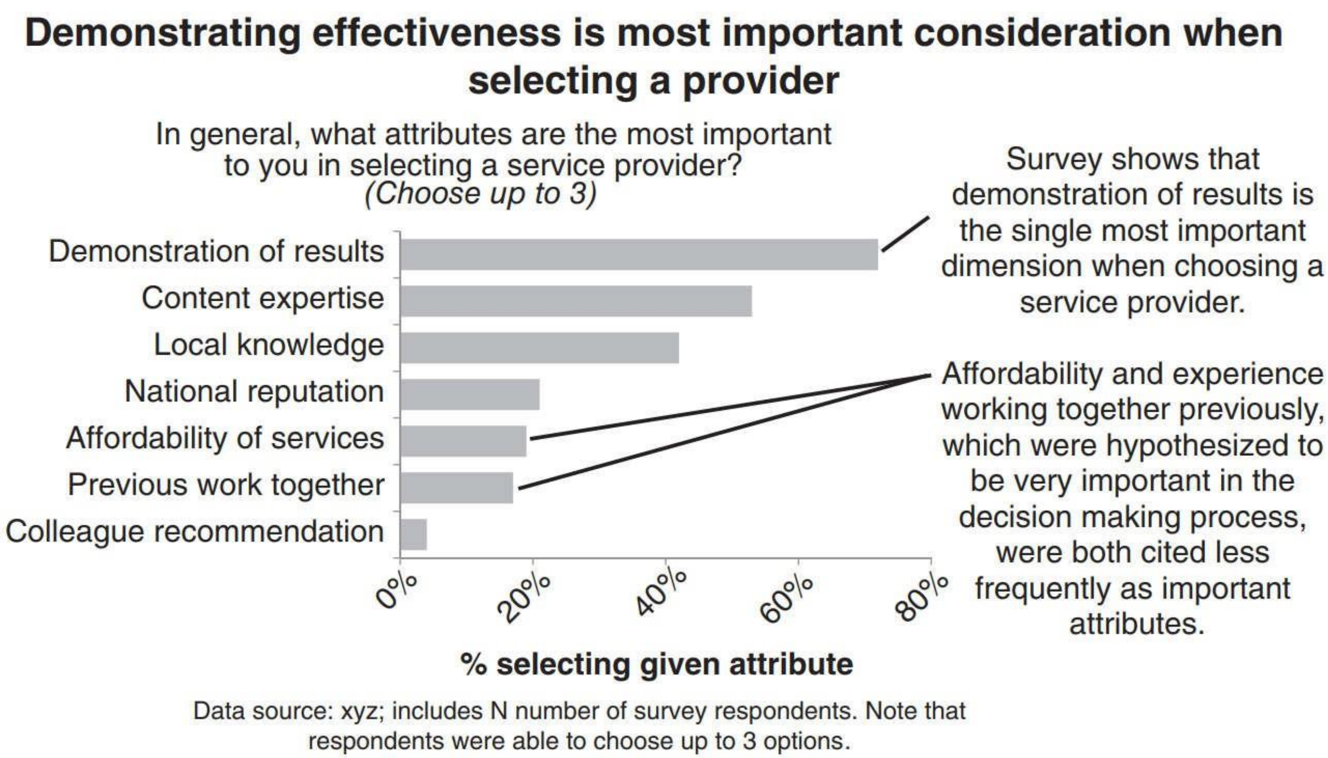
Connection + Similarity

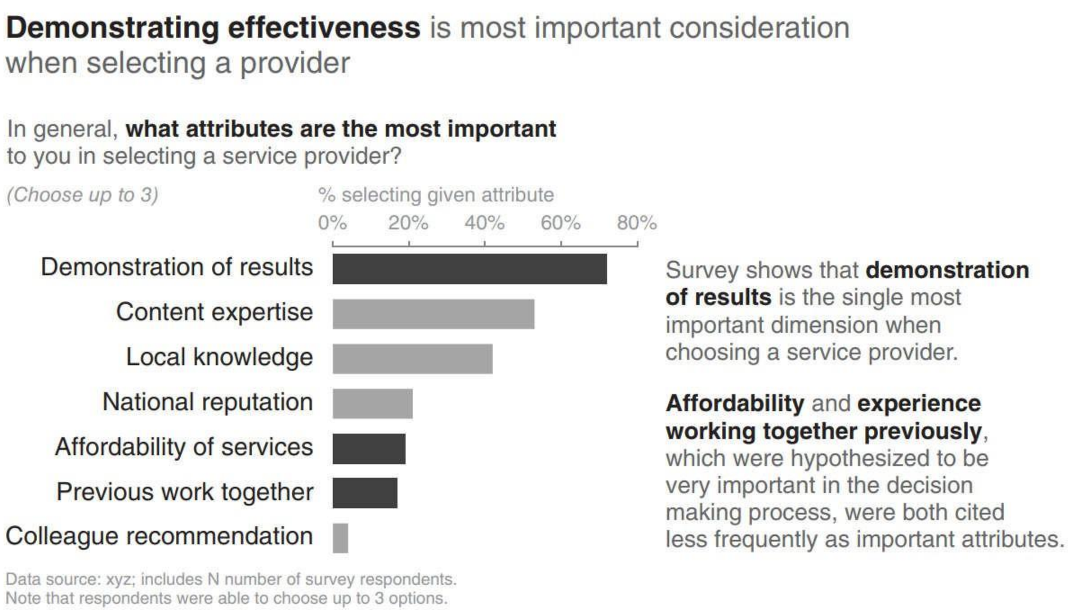


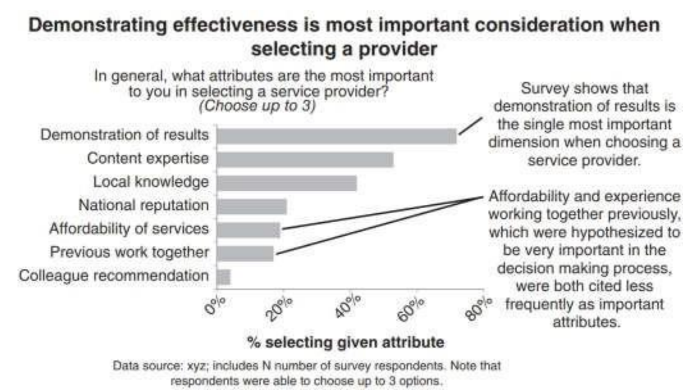
Other Types of Visual Clutter

Lack of visual order



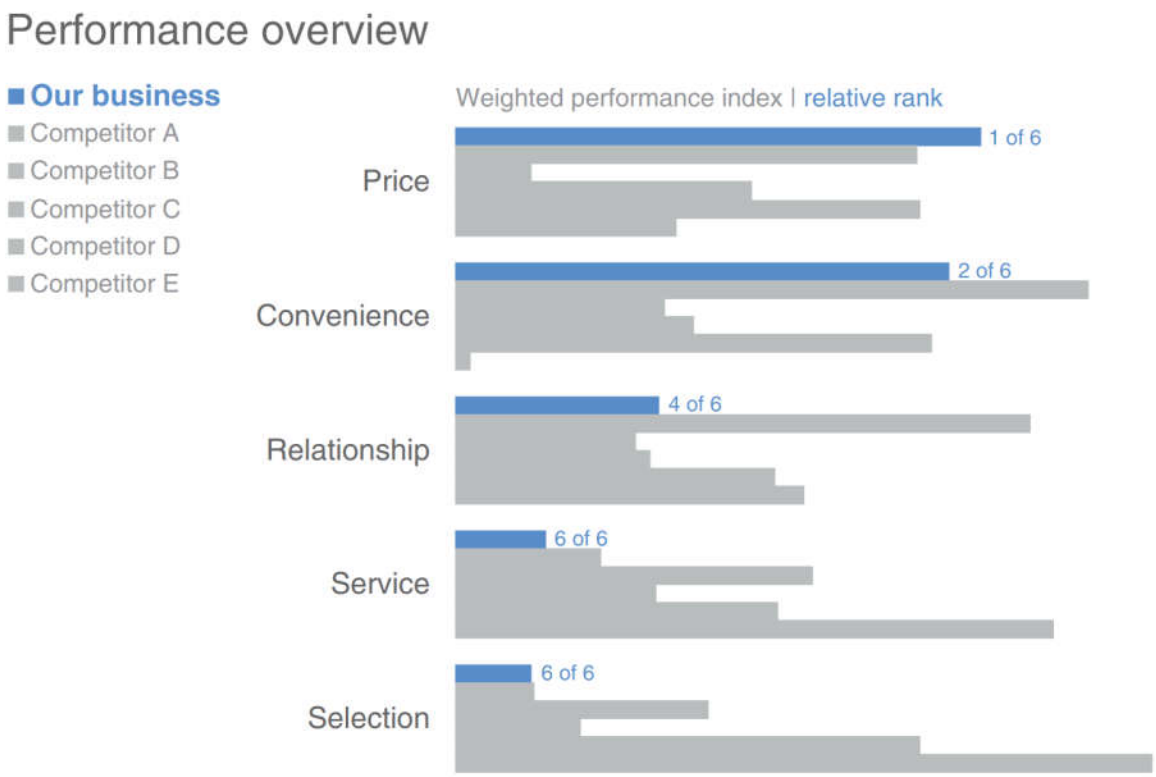
Creating visual order





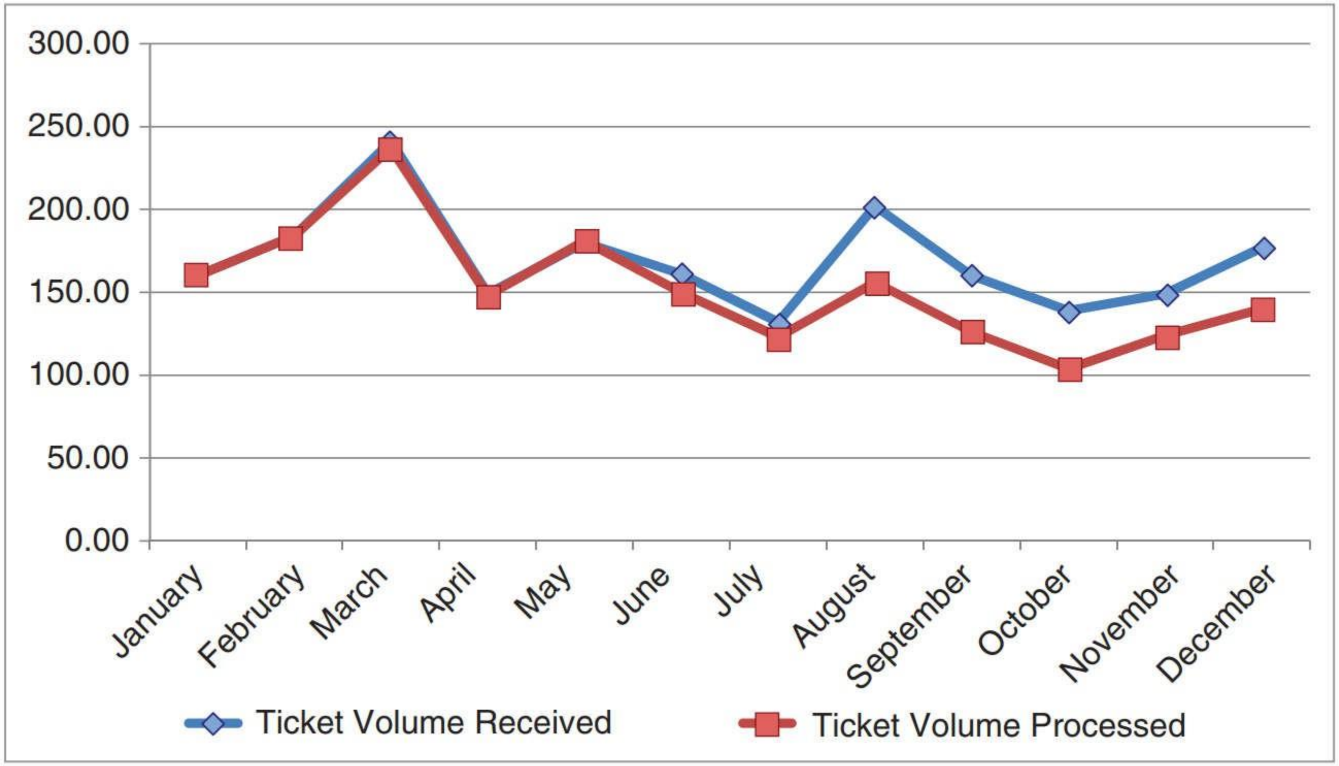
Non-strategic use of contrast

Strategic use of contrast



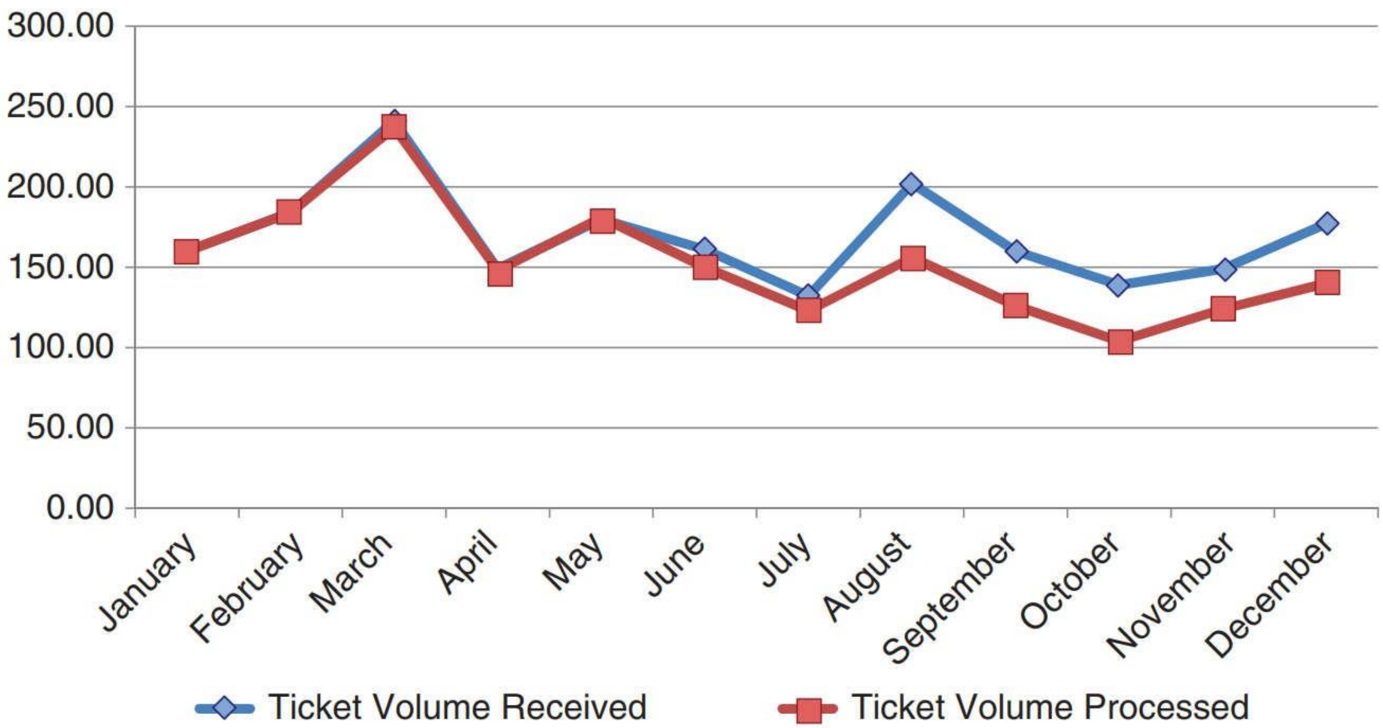
Decluttering: step-by-step (Example 1)

Example

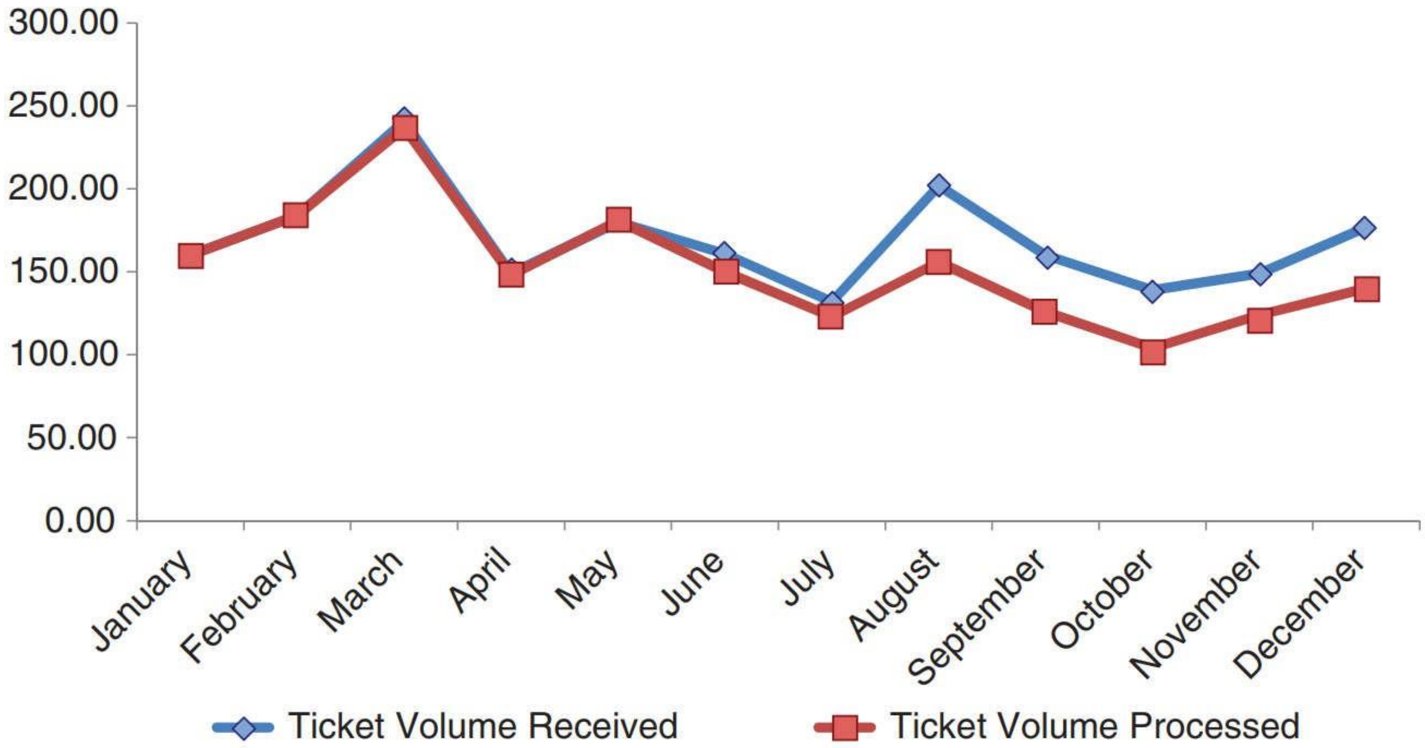


300.00
250.00
200.00
150.00
100.00
50.00
0.00
January
February
March
April
May
June
July
August
September
October
November
December
Ticket Volume Received
Ticket Volume Processed

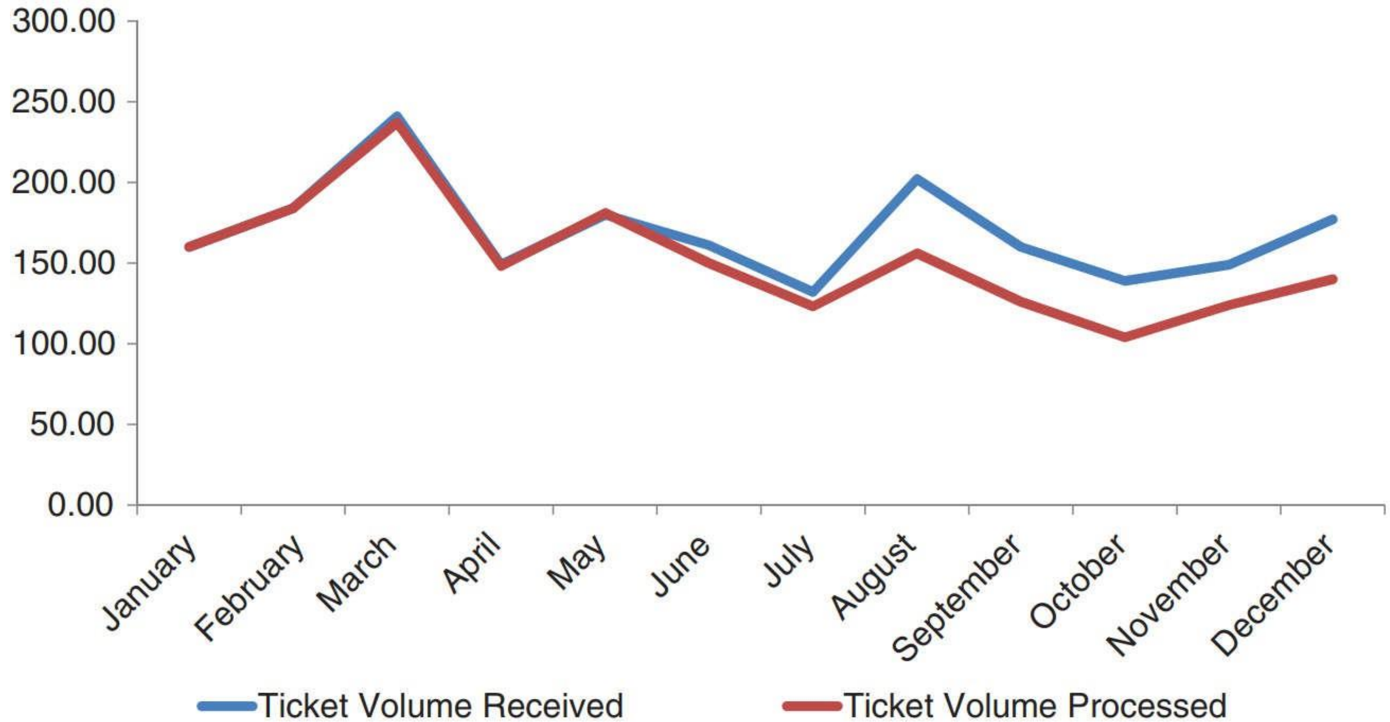
1. Remove chart border



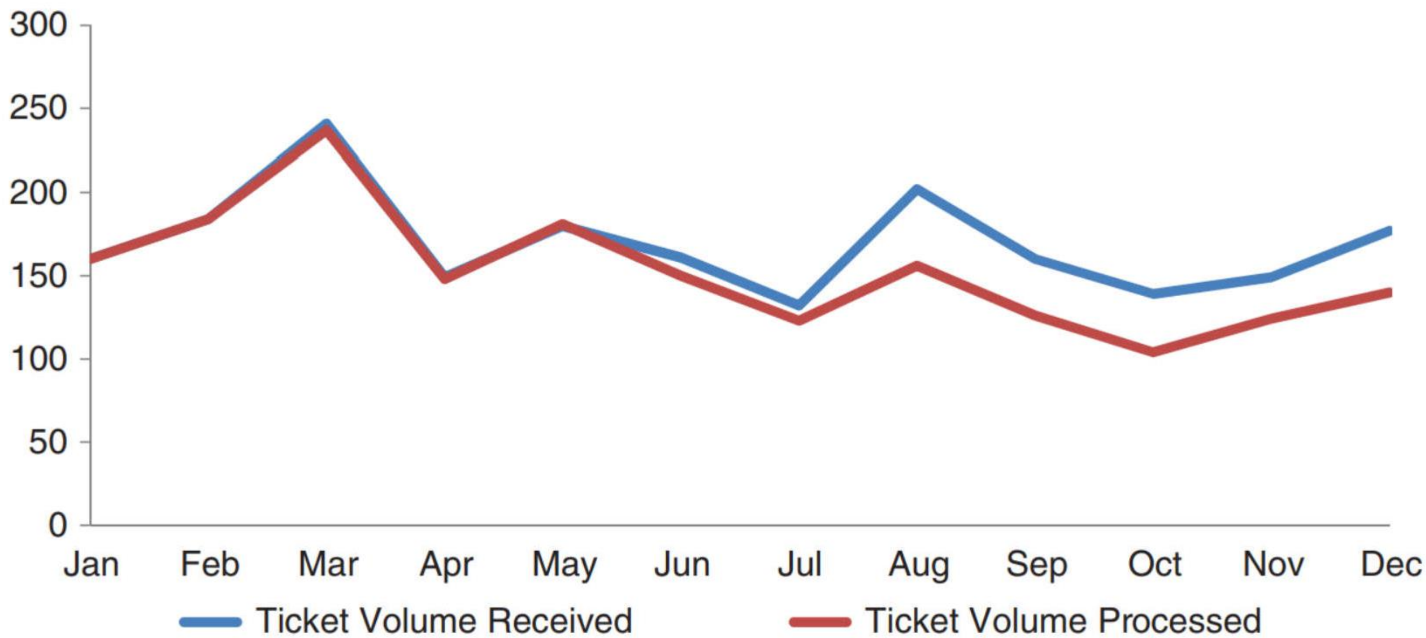
2. Remove gridlines



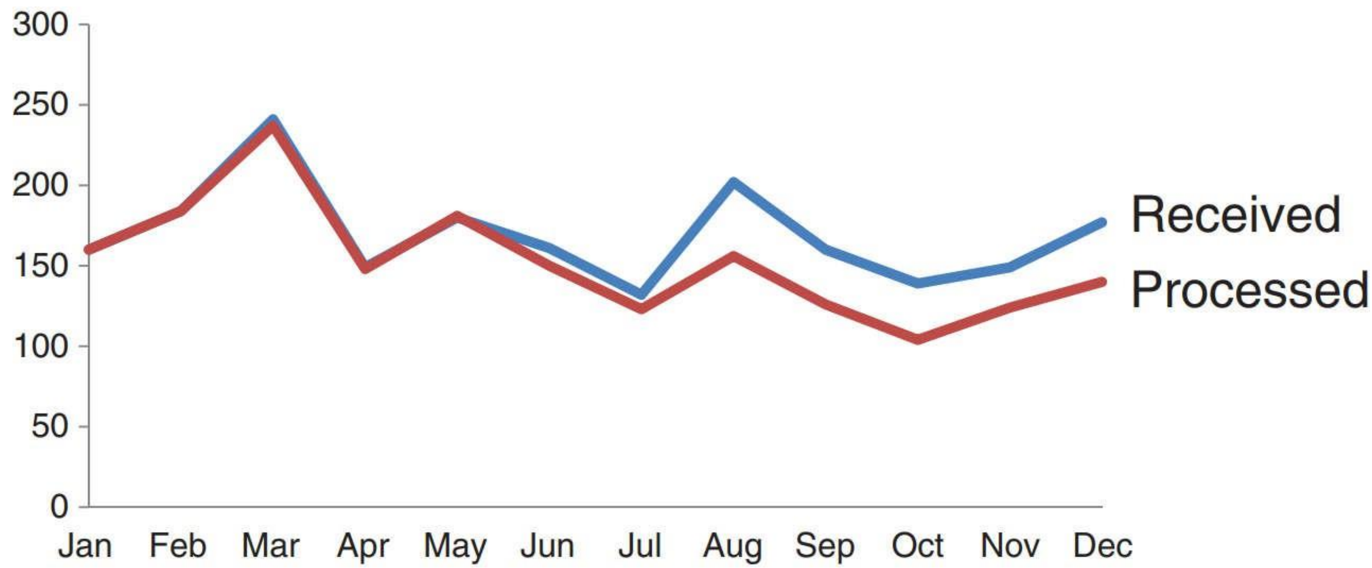
3. Remove data markers



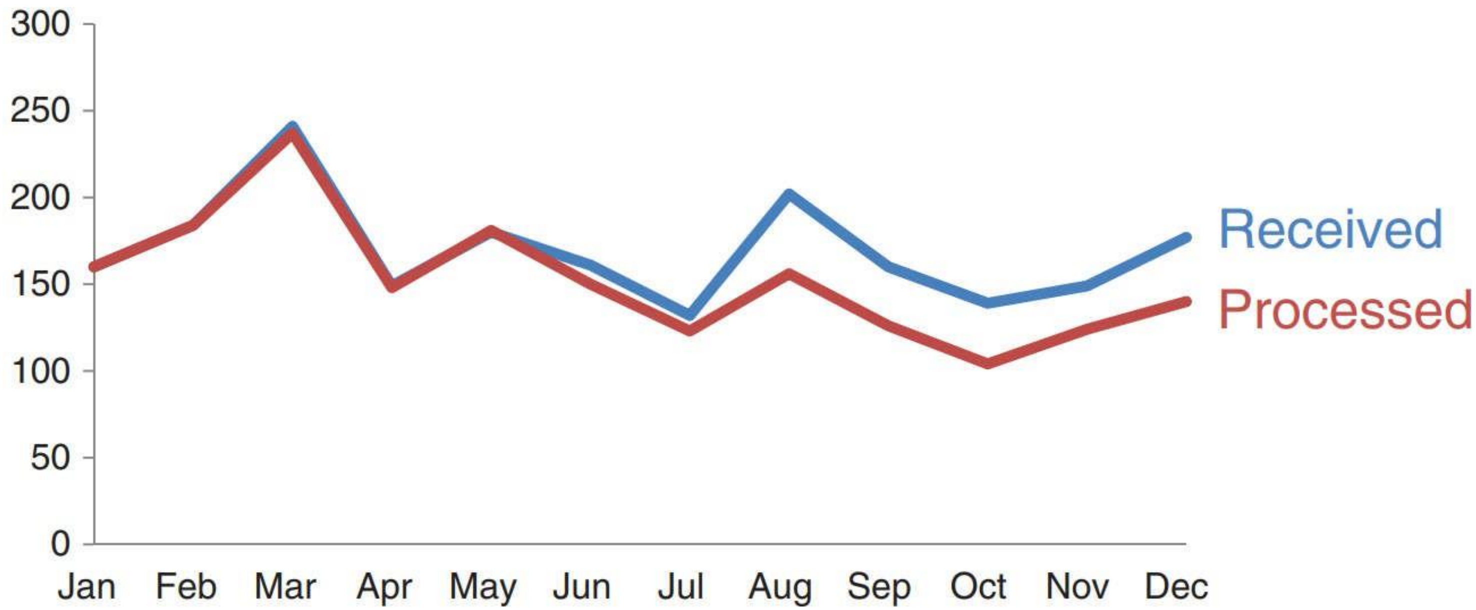
4. Clean up axis labels



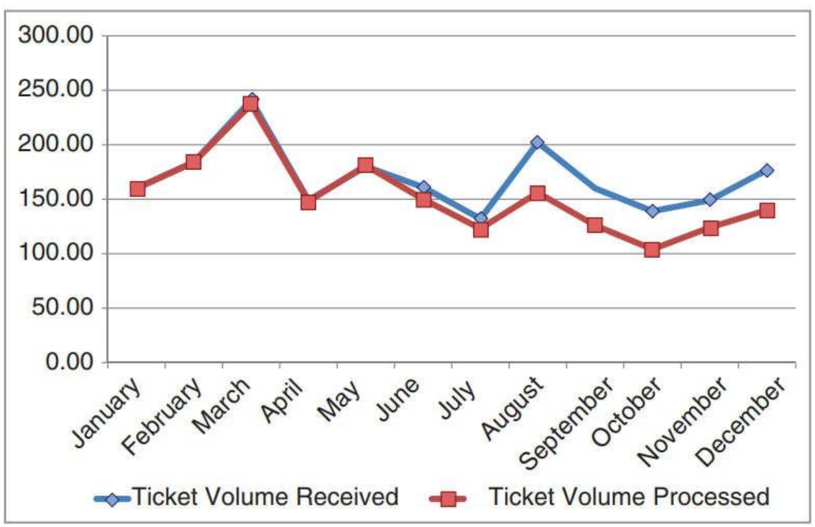
5. Label data directly

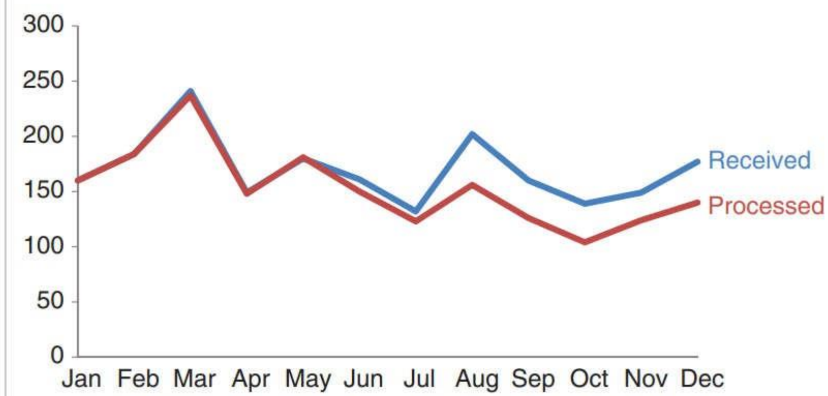


6. Leverage consistent color



Decluttered Graph





Decluttering: step-by-step (Example 2)

Example

Time to Close Deal
Goal = 90 days
140.0
120.0
100.0
80.0
60.0
40.0
20.0
0.0
88.2
82.2
76.3
71.4
47.8
88.7
76.1
81.0
71.4
88.4
58.6
120.2
79.9
83.5
69.4
73.8
53.9
98.0
80.8
85.1
74.3
94.4
46.4
64.8
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Direct Sales
Indirect Sales

Remove heavy lines

Remove gridlines

Drop trailing zeros from y-axis labels

Eliminate diagonal text on x-axis

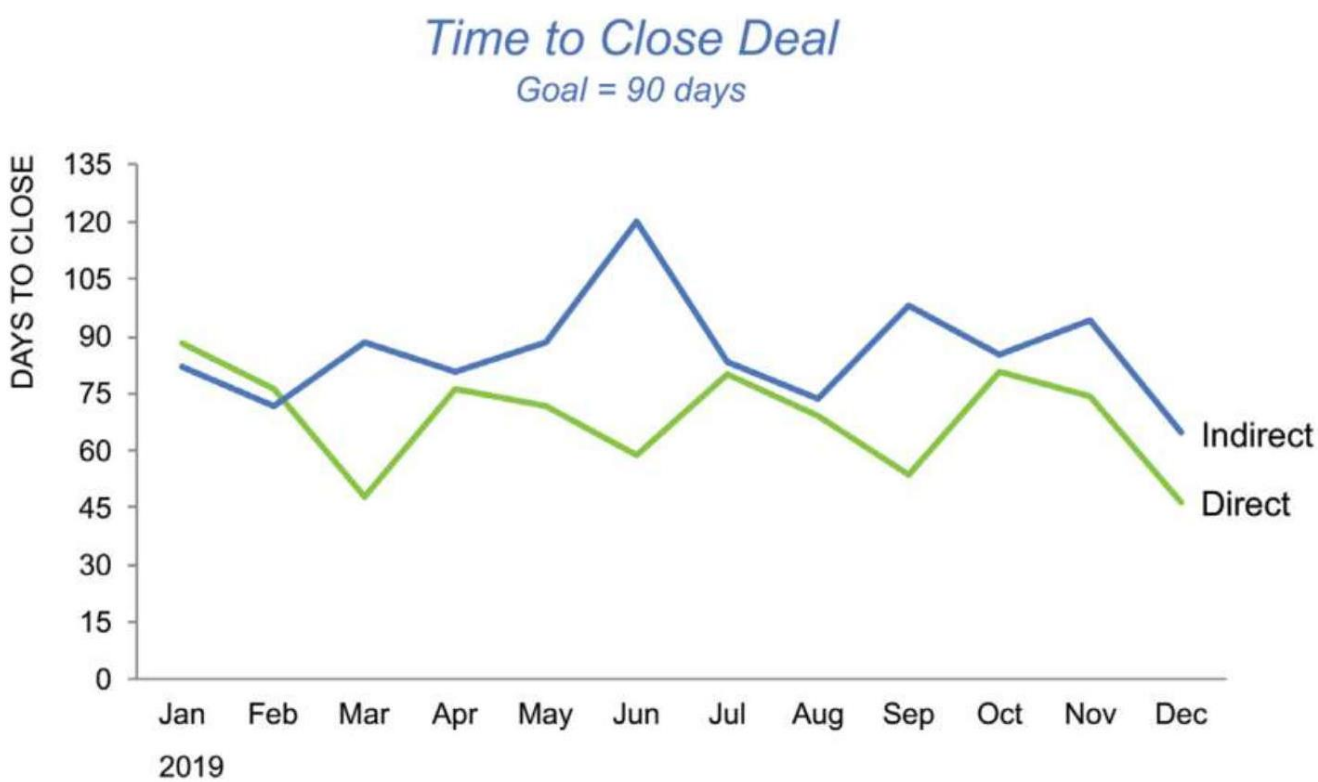
Thicken the bars

Pull data labels into ends of bars

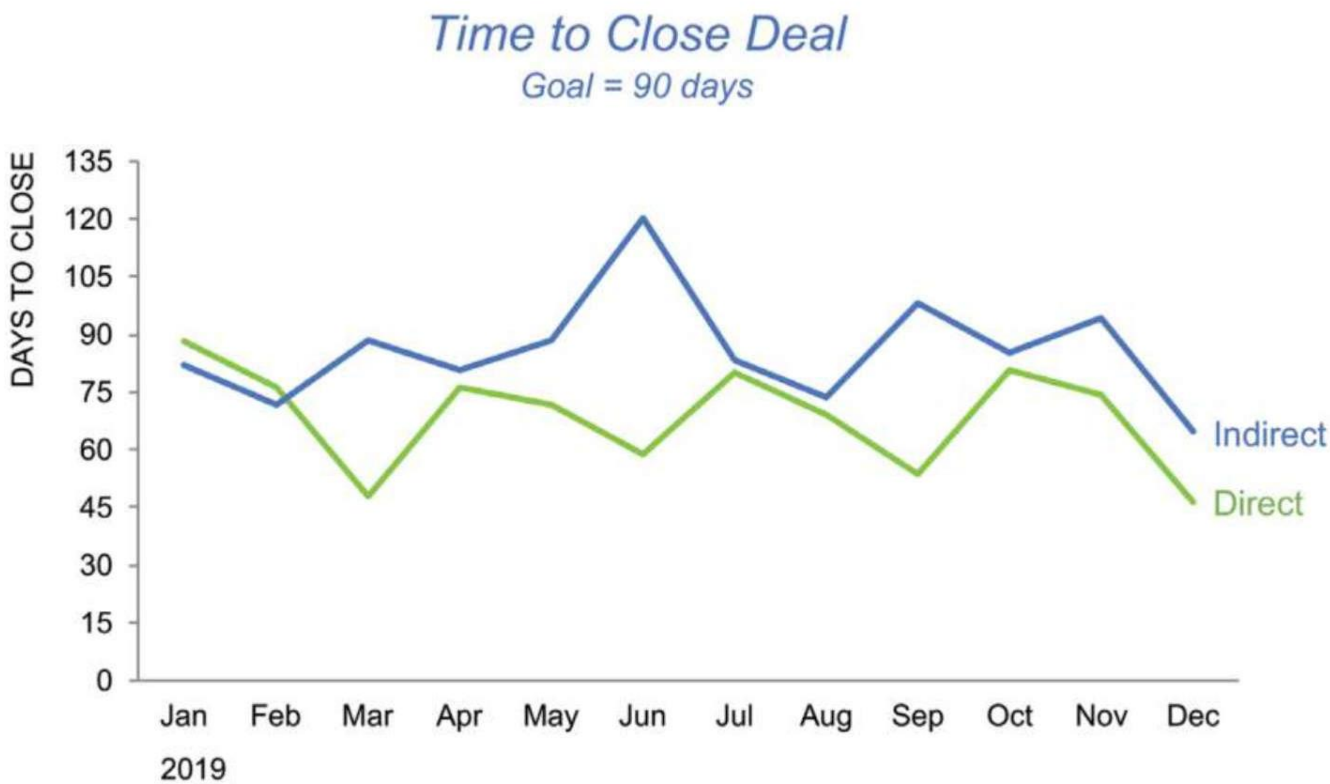
Eliminate data labels

Make it a line graph

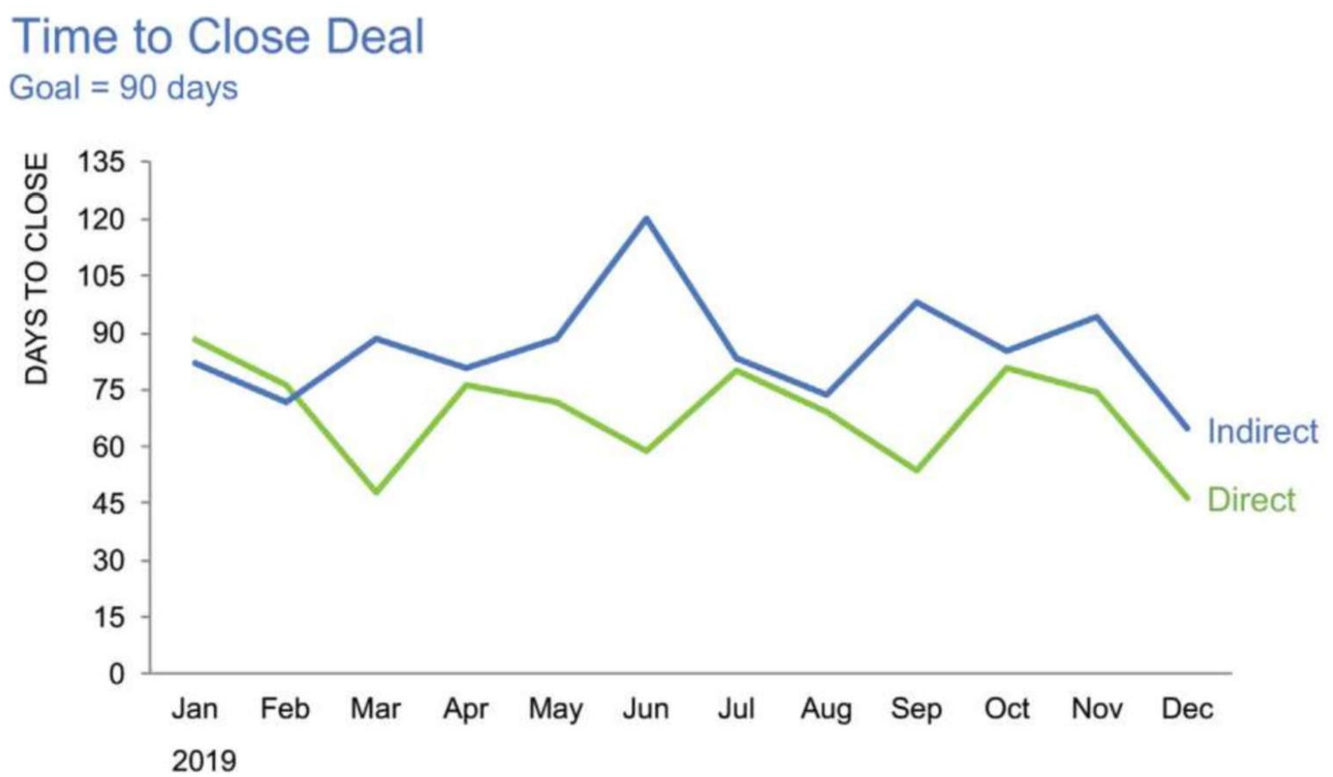
Label the data directly



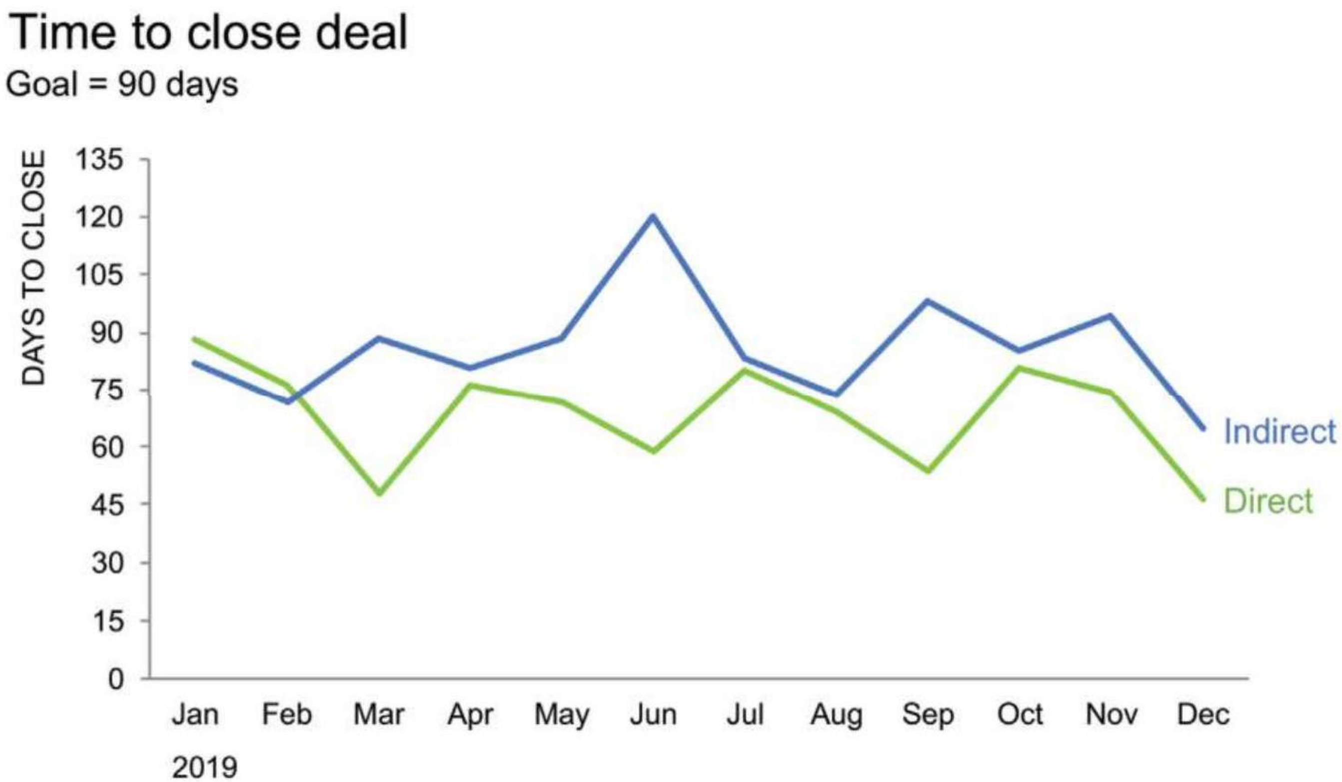
Make data labels the same color as the data



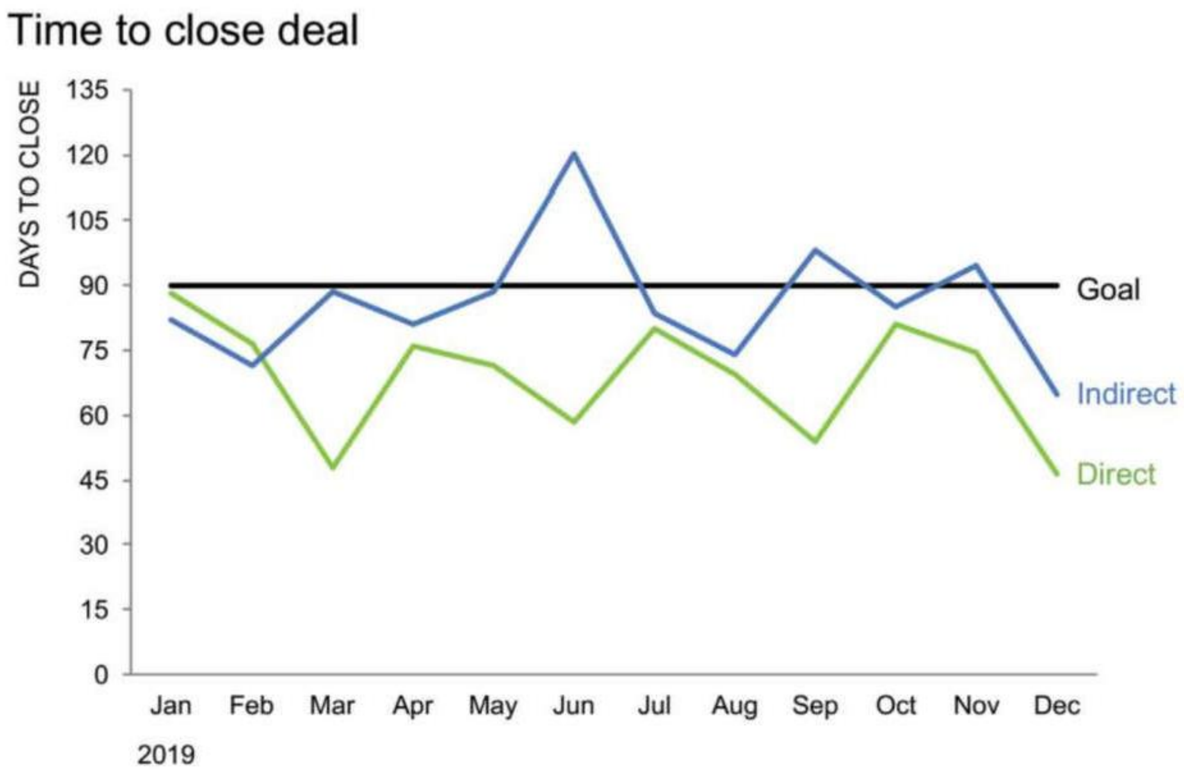
Upper-left-most orient graph title



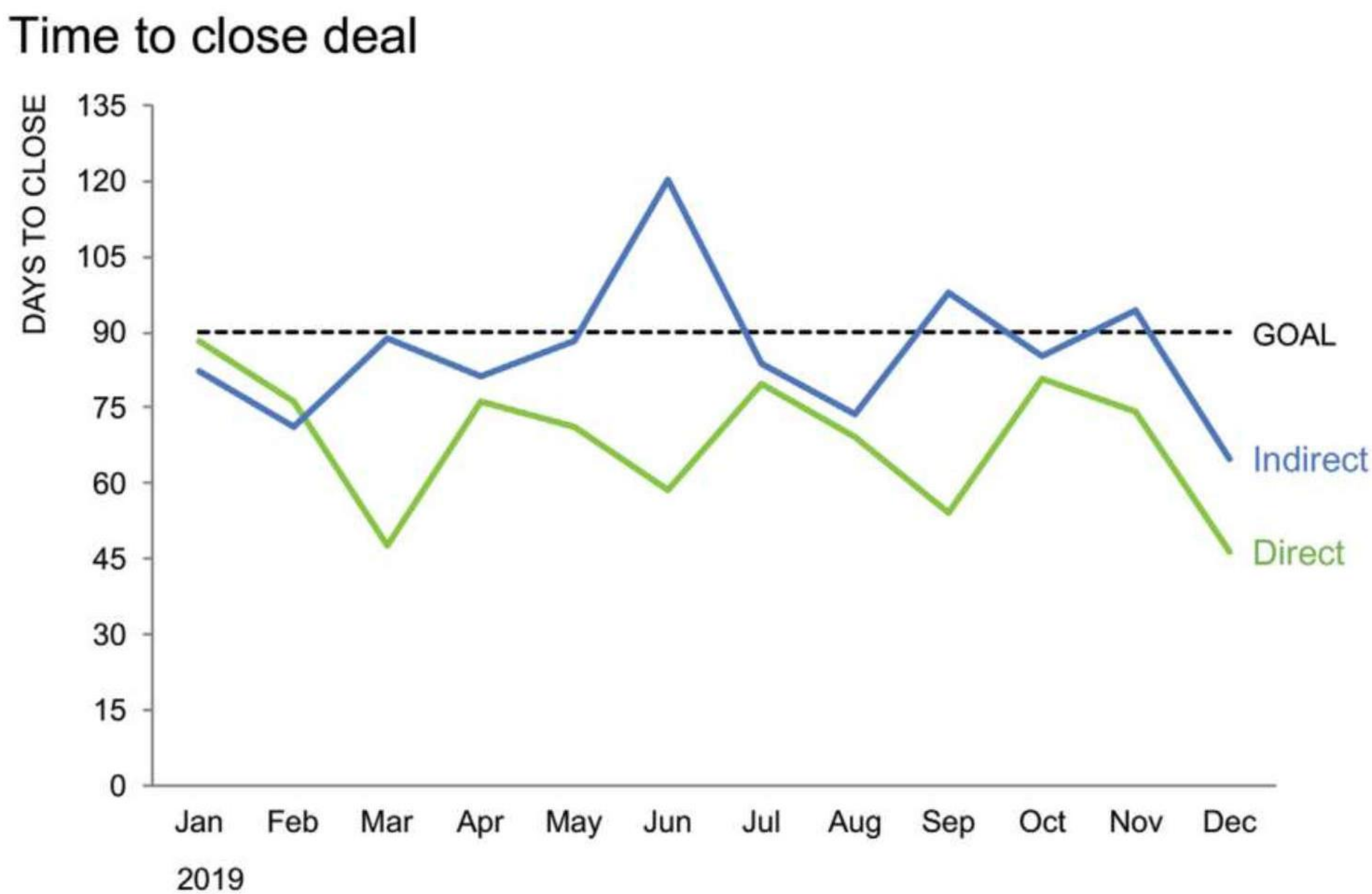
Remove title color



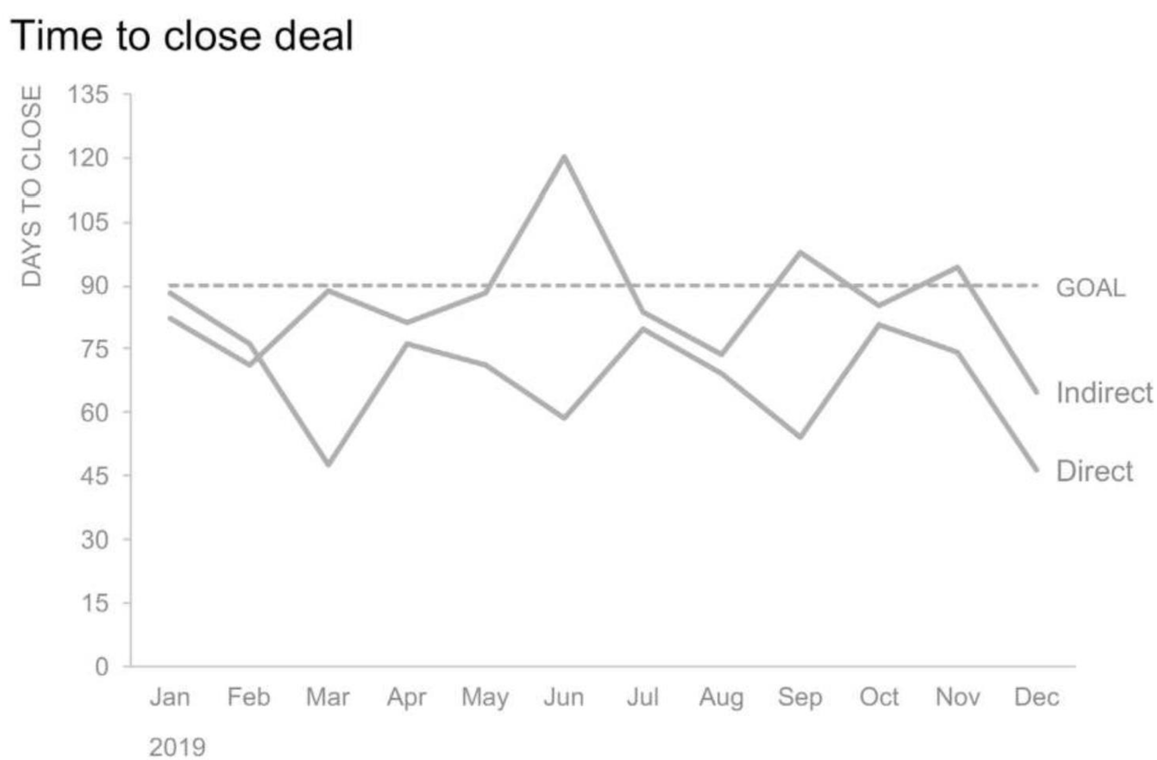
Put the goal in the graph



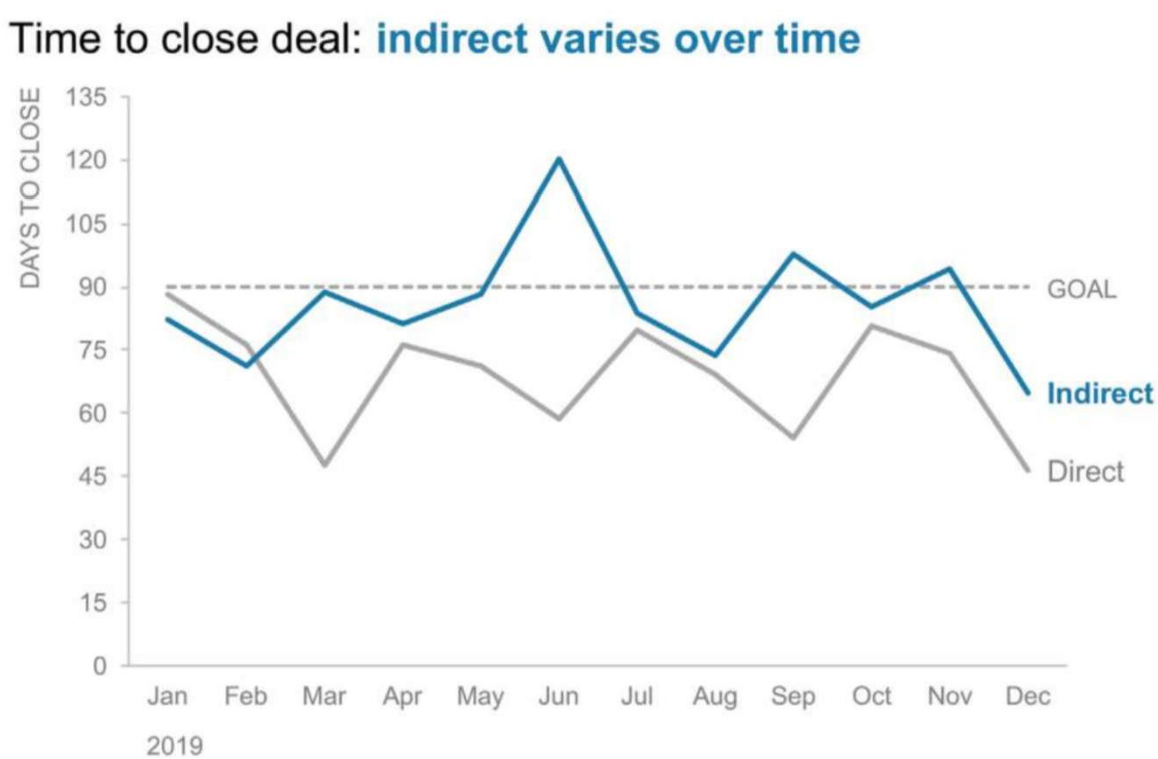
Iterate to best visualize *Goal*


Time to close deal
DAYS TO CLOSE
135
120
105
90
75
60
45
30
15
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2019
GOAL
Indirect
Direct

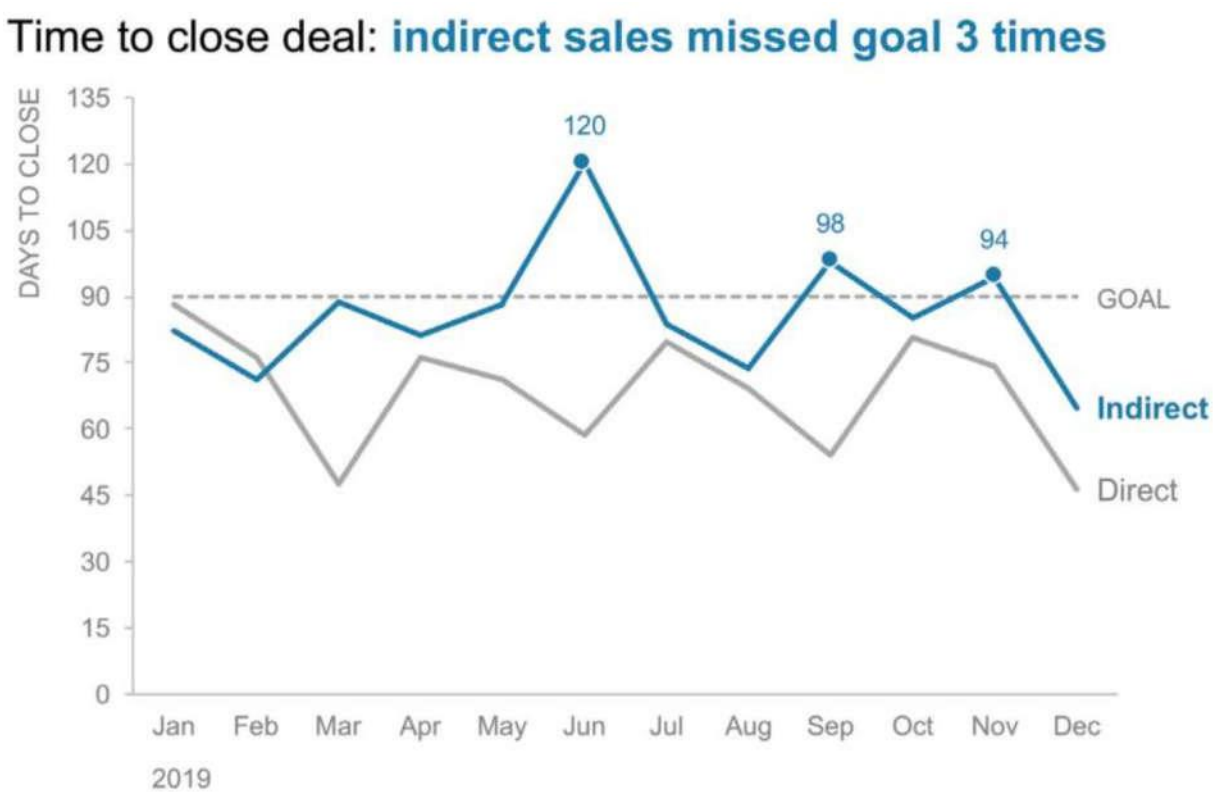
Remove color



Focus attention



Focus attention elsewhere



Introduce more color to direct attention

Compare!


Time to close deal: indirect sales missed goal 3 times
DAYS TO CLOSE
135
120
105
90
75
60
45
30
15
0
120
98
94
GOAL
Indirect
Direct
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2019


Time to Close Deal
Goal = 90 days
140.0
120.0
100.0
80.0
60.0
40.0
20.0
0.0
88.2
82.2
76.3
71.4
47.8
88.7
76.1
81.0
71.4
88.4
58.6
120.2
79.9
83.5
69.4
73.8
53.9
98.0
80.8
85.1
74.3
94.4
46.4
64.8
Jan-19
Feb-19
Mar-19
Apr-19
May-19
Jun-19
Jul-19
Aug-19
Sep-19
Oct-19
Nov-19
Dec-19
Direct Sales
Indirect Sales